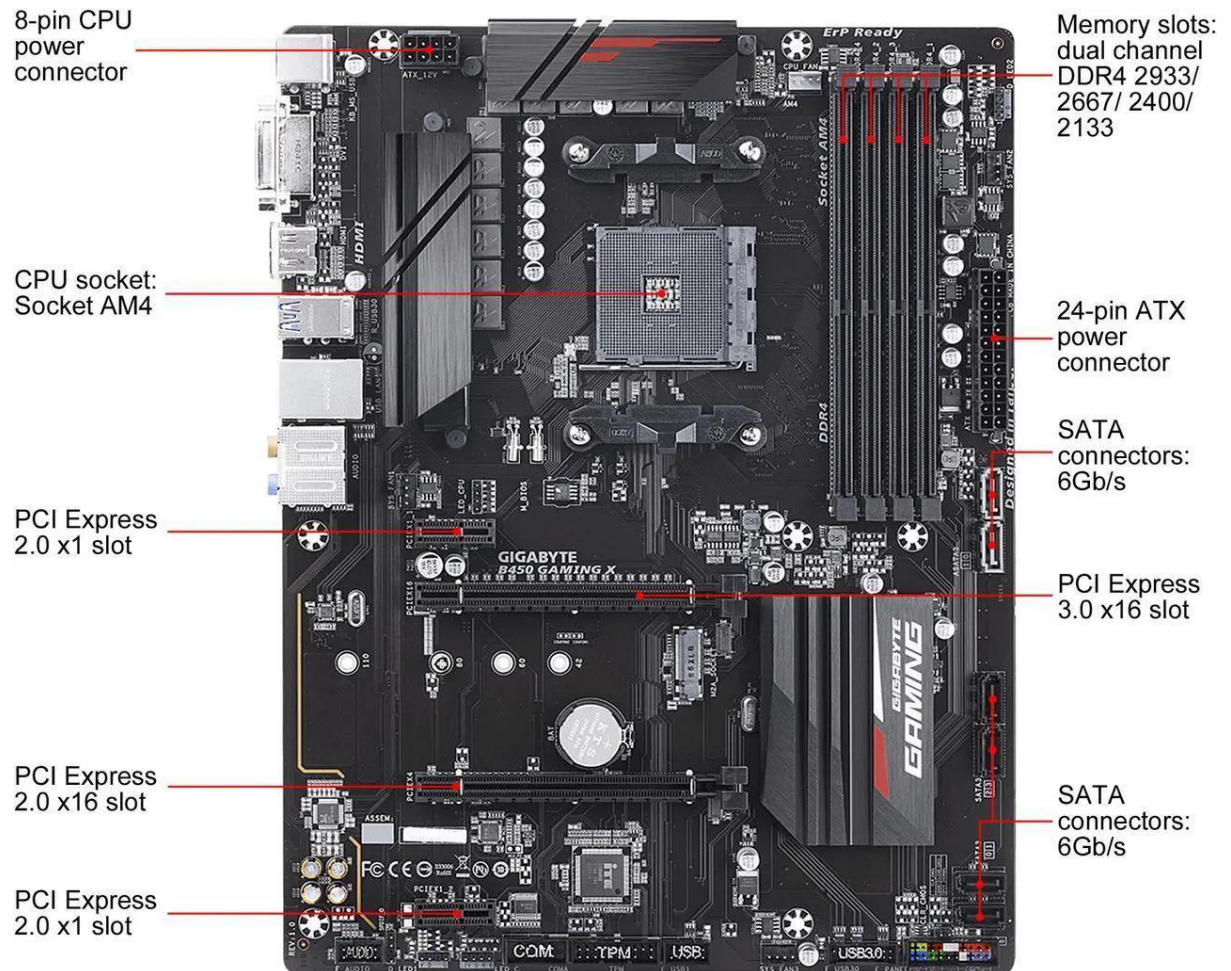


MOTHERBOARD AND CPU

1. Take a clear photo of the motherboard inside your own or a friend's laptop/desktop (or find a labeled image online), and label the CPU socket, RAM slots, PCIe slots, and power connectors using any image editing tool.

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Labels to include in your assignment

Component	Where it is located	Function
CPU Socket	Center/top-center of motherboard	Holds the processor (CPU)
RAM Slots (DIMM)	Usually beside the CPU socket	Holds RAM memory modules
PCIe Slots	Lower/center portion of motherboard	Used for GPU and other expansion cards
24-Pin Power Connector	Right edge of motherboard	Provides main power to the motherboard
8-Pin CPU Power Connector	Near the CPU socket, usually top-left/top edge	Provides power to the CPU

- Pick any three popular CPUs used in gaming laptops or desktops (for example, Intel Core i5-12400F, AMD Ryzen 5 5600, Intel Core i7-13700K). Research and create a table comparing their clock speed (GHz), number of cores/threads, cache size (L2/L3), TDP (W), and current price in India.

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CPU	Clock Speed	Cores / Threads	L2 Cache	L3 Cache	TDP	Current Price in India*
Intel Core i5-12400F	2.5 GHz base / 4.4 GHz turbo	6 / 12	7.5 MB	18 MB	65 W	≈ ₹11,000
AMD Ryzen 5 5600	3.5 GHz base / 4.4 GHz boost	6 / 12	3 MB	32 MB	65 W	≈ ₹12,684–₹14,799

CPU	Clock Speed	Cores / Threads	L2 Cache	L3 Cache	TDP	Current Price in India*
Intel Core i7-13700K	3.4 GHz P-core base / 5.4 GHz turbo	16 / 24	24 MB	30 MB	125 W	≈ ₹33,249

3. For each CPU in your comparison table, identify at least one compatible motherboard model (mention brand and model, e.g., ASUS PRIME B660M-A for Intel 12th Gen) and specify the socket type and form factor (ATX, mATX, ITX).

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CPU	Compatible Motherboard	Brand	Socket Type	Form Factor
Intel Core i5-12400F	ASUS PRIME B660M-A D4	ASUS	LGA1700	mATX
AMD Ryzen 5 5600	ASUS TUF Gaming B550-PLUS	ASUS	AM4	ATX
Intel Core i7-13700K	ASUS PRIME Z790-P	ASUS	LGA1700	ATX

1. Intel Core i5-12400F

ASUS PRIME B660M-A D4 is compatible with the 12th-generation Intel Core i5-12400F. It uses the **LGA1700 socket** and has an **mATX (Micro-ATX)** form factor. ASUS lists the board as supporting Intel 12th-, 13th-, and 14th-generation processors.

2. AMD Ryzen 5 5600

ASUS TUF Gaming B550-PLUS uses the **AM4 socket** and is an **ATX** motherboard. ASUS lists support for Ryzen 5000-series desktop processors, and its CPU support list specifically includes the Ryzen 5 5600.

3. Intel Core i7-13700K

ASUS PRIME Z790-P is compatible with the Intel Core i7-13700K. It uses the **LGA1700 socket** and is an **ATX** motherboard. ASUS specifies support for Intel 13th-generation Core processors.

4. Visit the official website of any one CPU you compared (Intel or AMD), and find the maximum supported RAM speed and PCIe version for that CPU. Write these details along with how they affect real-world performance in gaming or multitasking.

Hint: Look for 'Memory Types' and 'PCI Express Revision' in the CPU specs page.

->For this part, I selected the **Intel Core i5-12400F** from the previous comparison. I checked Intel's official processor specification page.

Specification	Intel Core i5-12400F
Maximum supported RAM	DDR5-4800 MT/s
Alternative RAM support	DDR4-3200 MT/s
PCIe version	PCIe 5.0 and PCIe 4.0
Maximum PCIe lanes	20 lanes

Intel's official specifications list support for up to **DDR5-4800 MT/s or DDR4-3200 MT/s**, depending on the memory type. Intel also specifies **PCI Express Revision 5.0 and 4.0**, with up to 20 PCIe lanes.

Effect on real-world performance

1. RAM speed — DDR5-4800 MT/s

Faster RAM can provide higher memory bandwidth, which can help the CPU access data more quickly. In gaming, this can sometimes improve frame rates and reduce performance bottlenecks, particularly in CPU-limited situations. For multitasking, faster memory can help when several applications are working with large amounts of data simultaneously.

However, the actual improvement depends on the motherboard, RAM configuration, game/application, and whether the system uses DDR4 or DDR5.

2. PCIe 5.0 / 4.0

PCIe is the connection standard used by devices such as graphics cards and NVMe SSDs. **PCIe 5.0 provides more bandwidth per lane than PCIe 4.0**, although the actual speed of a device is limited by the PCIe generation it supports.

